Worksheet 48 · Graph Transformations IGCSE

Edexcel IGCSE Mathematics · past paper questions

Name:	Subject: Math	Date:
Learning Objectives: Students will be able to: • Translate graphs using f(x + a) and f(x) + a • Reflect graphs using f(−x) and −f(x) • Identify the transformation from a graph or equation		
Grade: / 47	**Teacher's Feedback:**	

Answer in the space given. Show your working. Marks for each part are shown in brackets.

Key notes

#48 Graph Transformations

f(x)+a → Translation Up f(x)−a → Translation Down

f(x−a) → Translation Right f(x+a) → Translation Left

af(x) → Stretch in y axis by scale factor a

f(ax) → Stretch in x axis by scale factor $\frac{1}{a}$

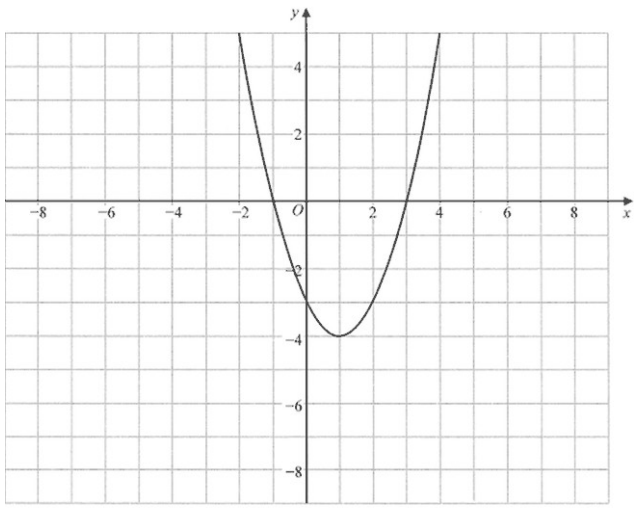
−f(x) → Reflection in x axis

f(−x) → Reflection in y axis

Questions *Graph Transformations*

Question 1 (2 marks)

The graph of $y = f(x)$ is shown on the grid.

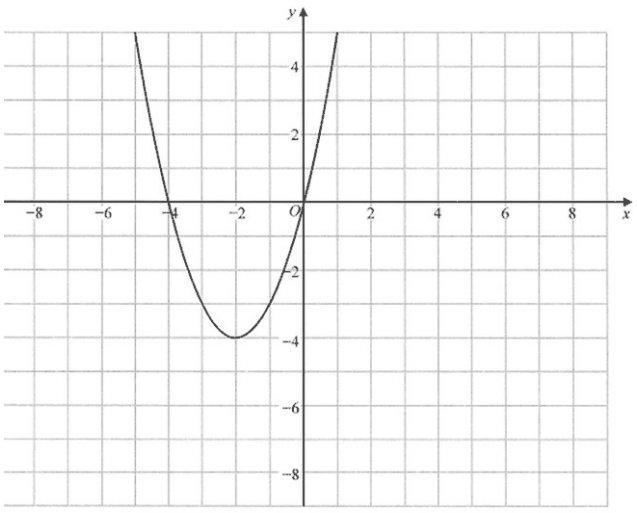


(a) On the grid above, sketch the graph of $y = f\left(\frac{1}{2}x\right)$

(2)

Question 2 (3 marks)

The graph of $y = f(x + k)$ is shown on the grid below.

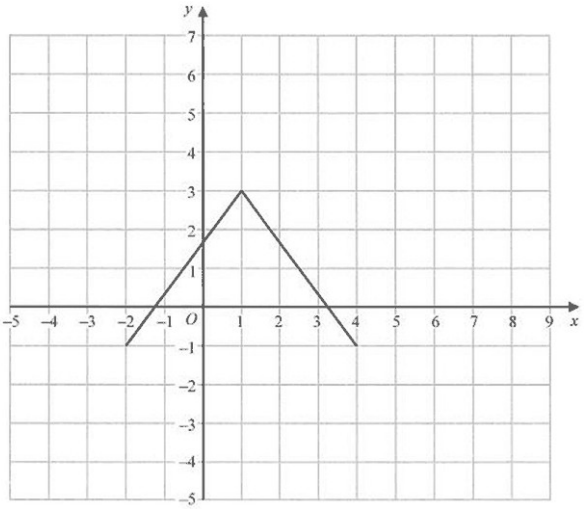


(b) Write down the value of k

(1)

Question 3 (2 marks)

Here is the graph of $y = f(x)$

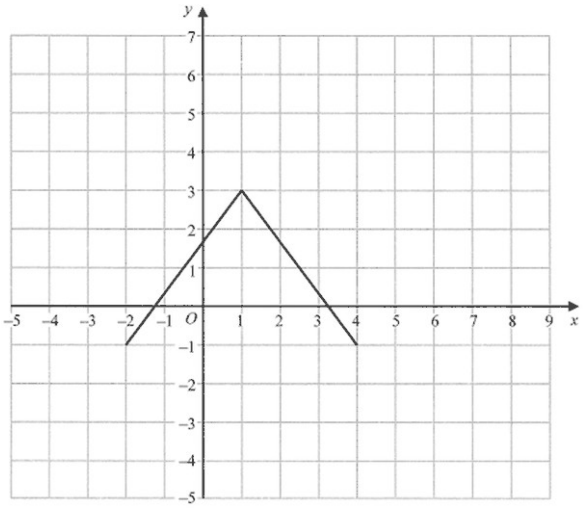


(a) On the grid above, draw the graph of $y = 2f(x)$

(2)

Question 4 (4 marks)

Here is the graph of $y = f(x)$



(b) On the grid above, draw the graph of $y = f(-x)$

(2)

Question 5 (2 marks)

The curve with equation $y = f(x)$ has one turning point.

The coordinates of this turning point are $(-6, -4)$

(a) Write down the coordinates of the turning point on the curve with equation

(i) $y = f(x) + 5$

(.....)

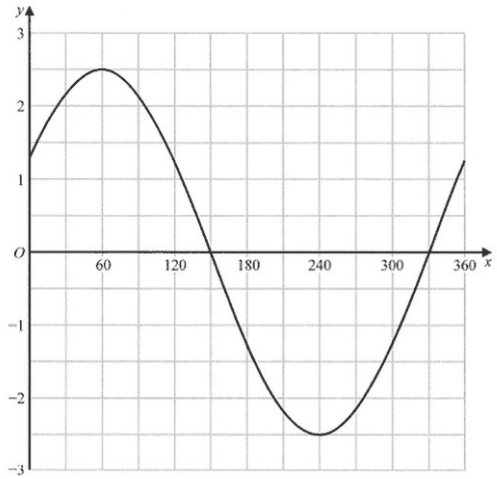
(ii) $y = f(3x)$

(.....)

(2)

Question 6 (4 marks)

The graph of $y = a\cos(x + b)°$ for $0 \leqslant x \leqslant 360$ is drawn on the grid.



(a) Find the value of a and the value of b.

$a =$ .....

$b =$ .....

(2)

Another curve C has equation $y = f(x)$
The coordinates of the minimum point of C are $(4, 5)$

(b) Write down the coordinates of the minimum point of the curve with equation

(i) $y = f(2x)$

(.....)

(ii) $y = f(x) - 7$

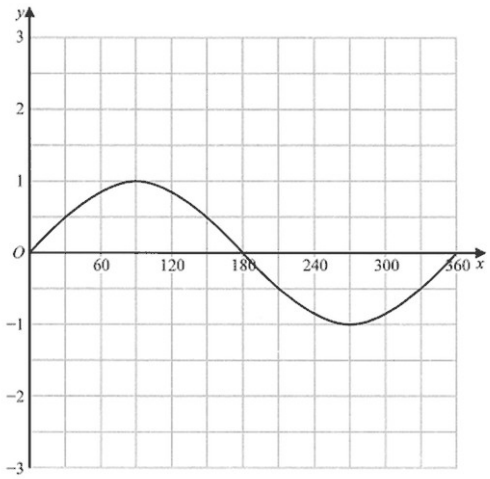
(.....)

(2)

(Total for Question 22 is 4 marks)

Question 7 (2 marks)

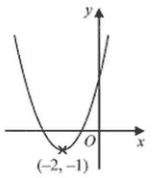
The graph of $y = \sin x°$ for $0 \leqslant x \leqslant 360$ is drawn on the grid.



(a) On the grid, draw the graph of $y = 2\sin(x + 30)°$ for $0 \leqslant x \leqslant 360$

(2)

Question 8 (5 marks)



The diagram shows the curve with equation $y = f(x)$
The coordinates of the minimum point of the curve are $(-2, -1)$

(a) Write down the coordinates of the minimum point of the curve with equation

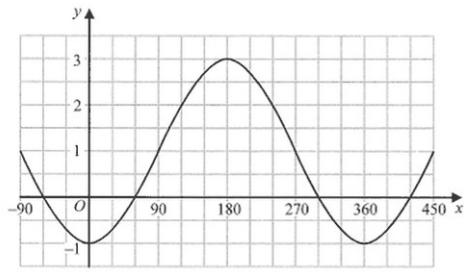
(i) $y = f(x - 5)$

(.....)

(ii) $y = \frac{1}{2} f(x)$

(.....)
(2)

The graph of $y = a\sin(x - b)° + c$ for $-90 \leqslant x \leqslant 450$ is drawn on the grid below.



(b) Find the value of a, the value of b and the value of c.

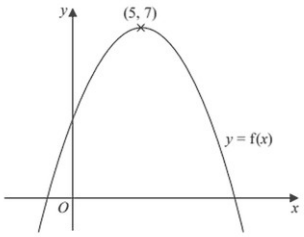
$a =$ .....

$b =$ .....

$c =$ .....
(3)

Question 9 (2 marks)

The diagram shows a sketch of the curve with equation $y = f(x)$



There is only one maximum point on the curve.
The coordinates of this maximum point are $(5, 7)$

Write down the coordinates of the maximum point on the curve with equation

(i) $y = f(x + 9)$

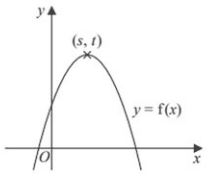
(.....)

(ii) $y = f(x) + 3$

(.....)

Question 10 (2 marks)

The diagram shows a sketch of part of the curve with equation $y = f(x)$



There is one maximum point on this curve.
The coordinates of this maximum point are (s, t)

Find, in terms of s and t, the coordinates of the maximum point on the curve with equation

(i) $y = f(x - 2)$

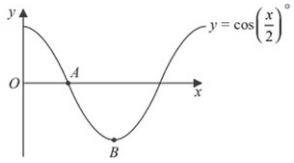
(.....)
(1)

(ii) $y = 3f(x)$

(.....)
(1)

Question 11 (2 marks)

The diagram shows a sketch of the graph of $y = \cos\left(\frac{x}{2}\right)°$



(i) Find the coordinates of the point A

(.....)
(1)

(ii) Find the coordinates of the point B

(.....)
(1)

Question 12 (3 marks)

The curve **C** has equation $y = f(x)$ where $f(x) = 9 - 3(x + 2)^2$
The point A is the maximum point on **C**.

(a) Write down the coordinates of A.

(.....)
(1)

The curve **C** is transformed to the curve **S** by a translation of $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$

(b) Find an equation for the curve **S**.

.....
(1)

The curve **C** is transformed to the curve **T**.
The curve **T** has equation $y = 3(x + 2)^2 - 9$

(c) Describe fully the transformation that maps curve **C** onto curve **T**.

.....
(1)

Question 13 (4 marks)

The diagram shows part of the curve with equation $y = f(x)$

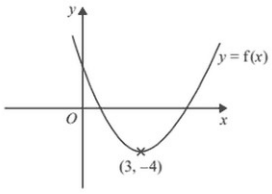


Diagram **NOT** accurately drawn

The coordinates of the minimum point on this curve are (3, –4)

(a) Write down the coordinates of the minimum point on the curve with equation

(i) $y = f(x - 4)$

(.....,.....)

(ii) $y = 3f(x)$

(.....,.....)

(iii) $y = f(\frac{1}{2}x)$

(.....,.....)

(3)

The curve with equation $y = f(x)$ is translated to give curve C.

C has a minimum at the point with coordinates (3, 5)
The equation of C is $y = f(x) + k$

(b) Write down the value of k

$k =$ .....

(1)

Question 14 (4 marks)

A curve has equation $y = f(x)$

The coordinates of the minimum point on this curve are (–9, 15)

(a) Write down the coordinates of the minimum point on the curve with equation

(i) $y = f(x + 3)$

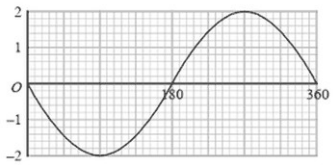
(.....,.....)

(ii) $y = \frac{1}{3}f(x)$

(.....,.....)

(2)

The graph of $y = a\cos(x + b)°$ for $0 \leqslant x \leqslant 360$ is drawn on the grid below.



Given that $a > 0$ and that $0 < b < 360$

(b) find the value of a and the value of b.

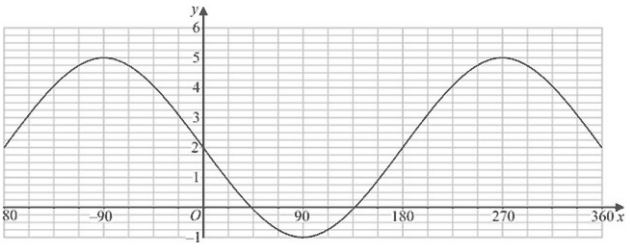
$a =$ .....

$b =$ .....

(2)

Question 15 (6 marks)

The graph of $y = a\cos(x - b)° + c$ for $-180 \leqslant x \leqslant 360$ is drawn on the grid below.



(d) Find the value of a, the value of b and the value of c.

$a =$ .....

$b =$ .....

$c =$ .....

(3)